SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
TIRUCHIRAPPALLI



LABORATORY REPORT

Subject:
Computer Networks Laboratory

Computer Networks LAB Test

B.Tech. Computer Science and Engineering
with Specialization in Cyber Security

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Submitted To:
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Lab 1: Introduction to Packet Tracer, Peer to Peer Communication, Study of Cables and its Color Codes

1. Aim

To get familiar with the Cisco Packet Tracer interface, to build a simple network of two computers and a switch, to assign IP addresses and test peer-to-peer communication with the ping command, and to study the Ethernet cables and the colour codes of their wires.

2. Software Required

Cisco Packet Tracer, installed on a personal computer running Windows.

3. Theory

Cisco Packet Tracer is a network simulation tool that lets us design a network, configure the devices and test how they communicate without using any real hardware. Its main window has a workspace in the middle where the network is drawn, a device selection box at the bottom to pick end devices, switches, routers and cables, and a toolbar with the basic editing tools.

In peer-to-peer communication two or more computers exchange data directly with each other and no device acts as a dedicated server. Every computer on the network needs a unique IP address and a subnet mask, and computers that are to talk to each other must be in the same network. The ping command sends ICMP echo request packets to another device and waits for the replies, so it is the easiest way to check whether two computers are connected.

A straight-through cable has the same wire order (T568B) at both ends and is used to join different types of devices, for example a computer to a switch. A crossover cable has the T568A order at one end and the T568B order at the other, so that the transmit and receive pairs are swapped. It is used to join similar devices, for example a computer to a computer.

4. Procedure

1. Open Packet Tracer:
    - Launch Cisco Packet Tracer on the computer.
    - Familiarize yourself with the interface, including the workspace, device selection and tools.
2. Create a simple network:
    - Drag two computers (PC-PT) onto the workspace.

- Drag a switch (Switch-PT) onto the workspace.
- Connect each computer to the switch using straight-through Ethernet cables.

3. Configure IP addresses:
    - Click on the first computer, go to the Desktop tab and select IP Configuration.
    - Assign an IP address (192.168.10.1) and a subnet mask (255.255.255.0).
    - Click on the second computer, go to the Desktop tab and select IP Configuration.
    - Assign an IP address (192.168.10.2) and a subnet mask (255.255.255.0).
4. Test peer-to-peer communication:
    - On the first computer, open the Command Prompt from the Desktop tab.
    - Use the ping command to test connectivity to the second computer (ping 192.168.10.2).
    - Observe the response to ensure that the computers can communicate.
5. Study cables and colour codes:
    - Examine the different types of network cables (Ethernet, crossover).
    - Note the colour codes of each wire in the cables.
    - Straight-through cable: used to connect different devices like a computer to a switch.
    - Crossover cable: used to connect similar devices like a computer to a computer.

5. Implementation and Output

5.1 Creating the Network

Two PCs (PC0 and PC1) and one switch were dragged onto the workspace. Each PC was joined to the switch with a straight-through Ethernet cable, using the FastEthernet port of the PC and a free port on the switch. The link lights turned green after a few seconds, which shows that the connections are working.

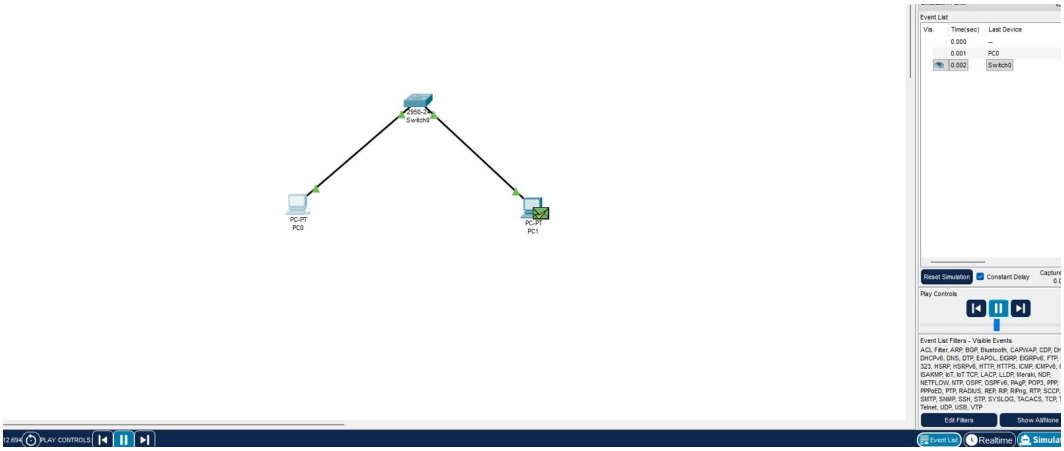


Fig. 1: Network of two PCs connected to a switch

5.2 Configuring the IP Addresses

By clicking on PC0 and opening Desktop, then IP Configuration, the static option was selected. The IP address 192.168.10.1 and the subnet mask 255.255.255.0 were entered. The same steps were repeated on PC1 with the IP address 192.168.10.2 and the same subnet mask.

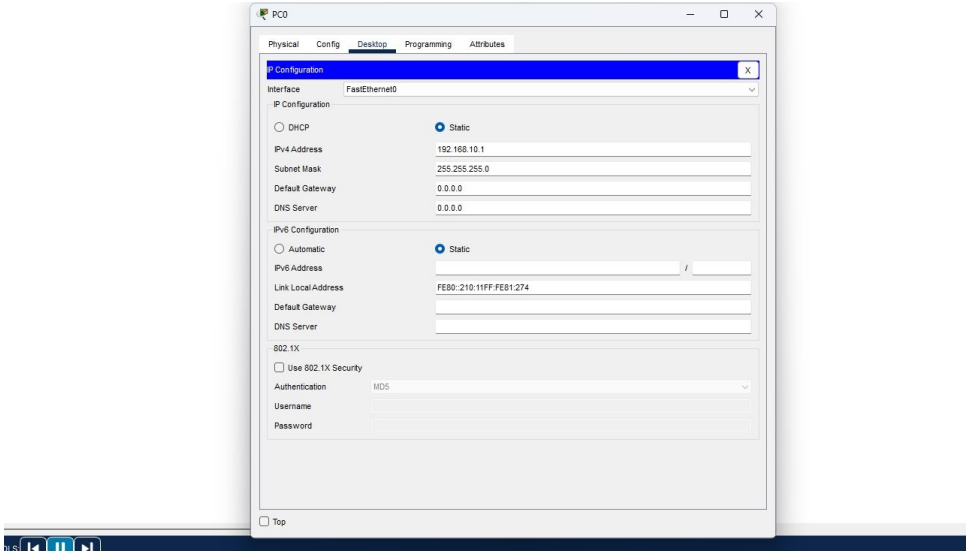


Fig. 2: IP configuration of PC0

5.3 Testing Peer to Peer Communication

From the Command Prompt of PC0 the command ping 192.168.10.2 was given. All four packets were sent and four replies were received with no packet loss, so PC0 and PC1 can communicate with each other through the switch.

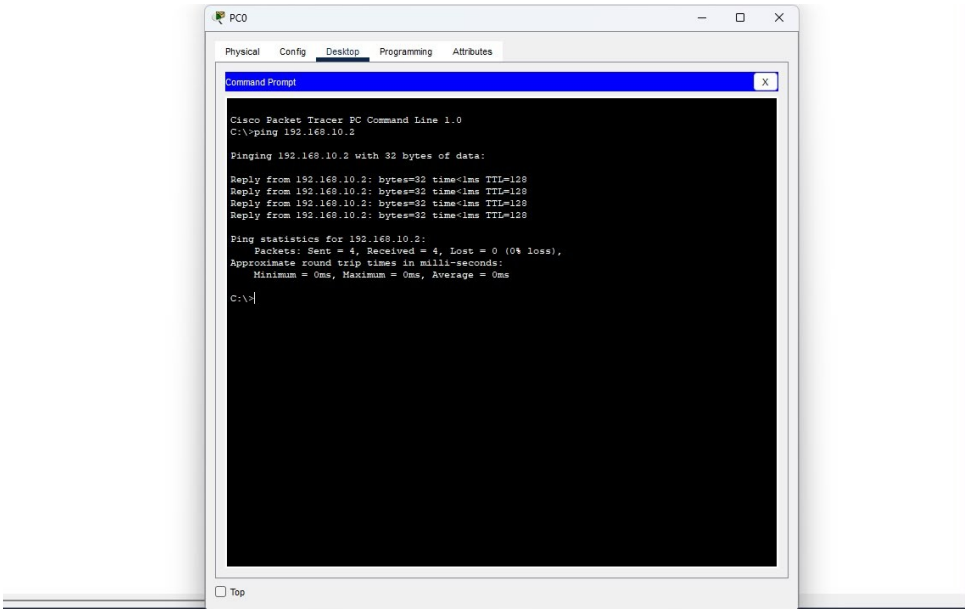


Fig. 3: Ping from PC0 to PC1

6. Study of Cables and Colour Codes

Ethernet cables used in networks have four twisted pairs, that is eight wires, which are placed in the RJ-45 connector in a fixed order. The two standard orders are T568A and T568B.

6.1 Straight through Cable

Both ends of a straight-through cable use the same standard, normally T568B, so pin 1 at one end goes to pin 1 at the other end and so on. It is used between unlike devices such as a PC and a switch, a PC and a hub, or a router and a switch.

Pin No.	**End 1 (T568B)**	**End 2 (T568B)**
1	White / Orange	White / Orange
2	Orange	Orange
3	White / Green	White / Green
4	Blue	Blue
5	White / Blue	White / Blue
6	Green	Green
7	White / Brown	White / Brown
8	Brown	Brown

6.2 Crossover Cable

A crossover cable has T568B at one end and T568A at the other. This swaps the transmit pair (pins 1 and 2) with the receive pair (pins 3 and 6), so the two devices can be connected directly. It is used between like devices such as a PC and a PC, or a switch and a switch.

Pin No.	**End 1 (T568B)**	**End 2 (T568A)**
1	White / Orange	White / Green
2	Orange	Green
3	White / Green	White / Orange
4	Blue	Blue
5	White / Blue	White / Blue
6	Green	Orange
7	White / Brown	White / Brown
8	Brown	Brown

6.3 Comparison

Point	Straight through	Crossover
Wiring	T568B at both ends	T568B at one end, T568A at the other
Connects	Different devices	Similar devices
Example	PC to switch	PC to PC

7. Result

Cisco Packet Tracer was explored and a network of two PCs and a switch was created with straight-through cables. After the IP addresses 192.168.10.1 and 192.168.10.2 were assigned, the ping command showed successful communication between the two PCs. The colour codes of straight-through and crossover cables were also studied.

8. Conclusion

In this lab I learned how to use the main parts of Packet Tracer, how to connect devices and give them IP addresses, and how to check connectivity with ping. I also understood why a straight-through cable is used between different devices and a crossover cable between similar devices, and how the T568A and T568B colour orders differ.

Lab 2: Implementation of Network Topologies

1. Aim

To implement bus, star, ring and mesh network topologies in Cisco Packet Tracer using three computers for each topology, assign IP addresses to the devices, and verify connectivity between all the computers with the ping command.

2. Software Required

Cisco Packet Tracer, installed on a personal computer running Windows.

3. Theory

A network topology is the way in which the devices of a network are physically or logically arranged and connected to each other. The choice of topology affects the cost of cabling, the speed of the network, how easily faults can be found and how the network behaves when one link or device fails.

In a bus topology all the devices share one common backbone cable. It needs very little cabling, but a break in the backbone stops the whole network. In a star topology every device is connected to a central switch, so a fault in one cable affects only that device, although the switch becomes a single point of failure. In a ring topology each device is connected to exactly two neighbours so that the connections form a closed loop and data travels around the ring. In a mesh topology every device is connected directly to every other device, which gives the most reliable network but needs the largest number of cables and ports.

4. Procedure

1. Open Packet Tracer: launch Cisco Packet Tracer on the computer.
2. Implement a Bus Topology:
    - Drag computers onto the workspace.
    - Connect them using a single backbone cable (coaxial cable).
3. Implement a Star Topology:
    - Drag computers and a switch onto the workspace.
    - Connect each computer to the switch using straight-through Ethernet cables.
4. Implement a Ring Topology:
    - Drag computers onto the workspace.

- Connect them in a circular manner using crossover cables.

5. Implement a Mesh Topology:
    - Drag computers onto the workspace.
    - Connect each computer to every other computer using crossover cables.
6. Test connectivity:
    - For each topology, assign IP addresses to the computers.
    - Use the ping command to test connectivity between all the computers.

5. Implementation and Output

5.1 Bus Topology

Three computers were placed in the workspace and joined to a single backbone line. IP addresses from the same network were then given to each computer under the Config tab. From the Command Prompt of one PC, the ping command was used to reach the other PCs, and every request received a reply.

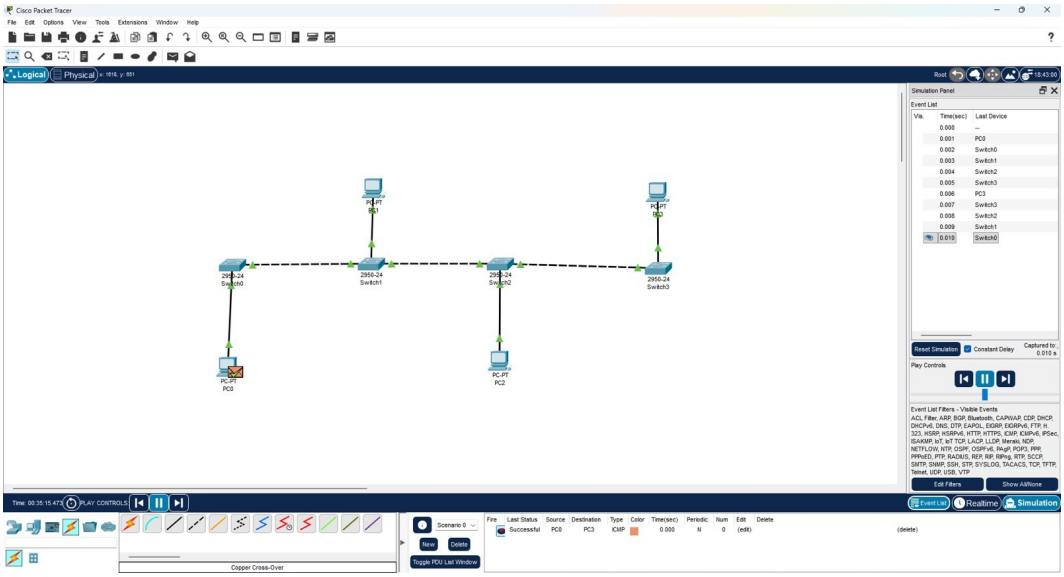


Fig. 1: Bus topology in Cisco Packet Tracer

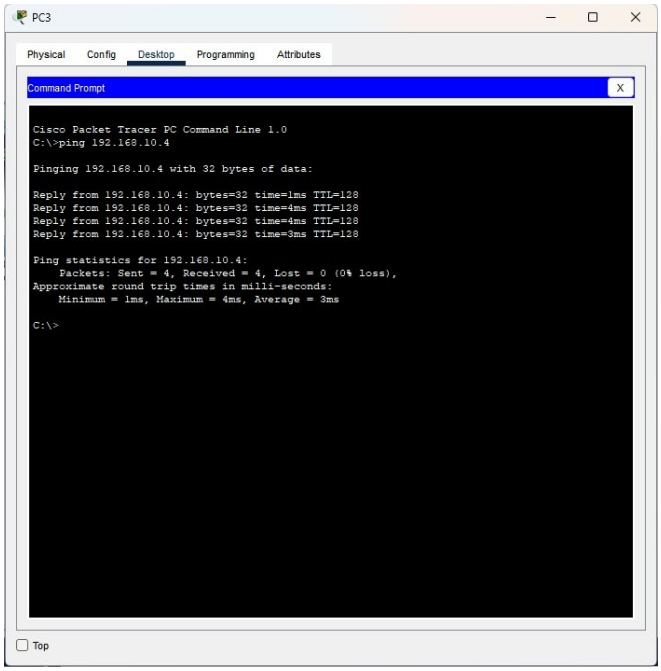


Fig. 2: Ping test between the PCs in the bus topology

5.2 Star Topology

A switch was placed at the centre and each computer was connected to it with a straight-through Ethernet cable. After the IP addresses were assigned, the PCs were pinged from one another through the switch, and all the pings were successful.

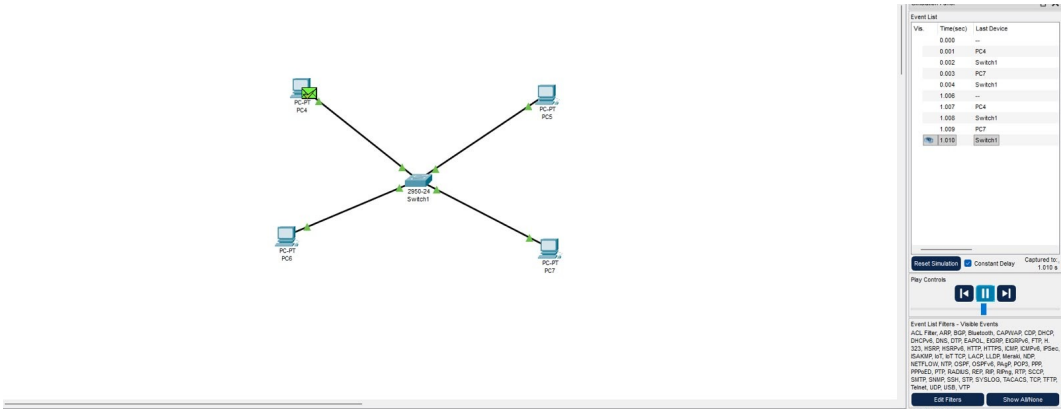


Fig. 3: Star topology in Cisco Packet Tracer

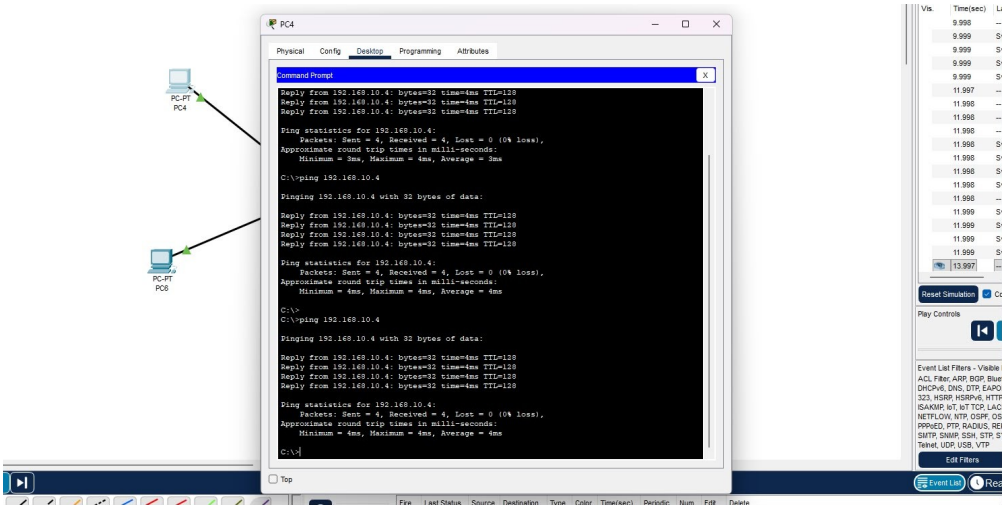


Fig. 4: Ping test between the PCs in the star topology

5.3 Ring Topology

The computers were connected one after another in a circle with crossover cables so that each device is linked to two neighbours. IP addresses were configured and connectivity was checked with the ping command from the Command Prompt.

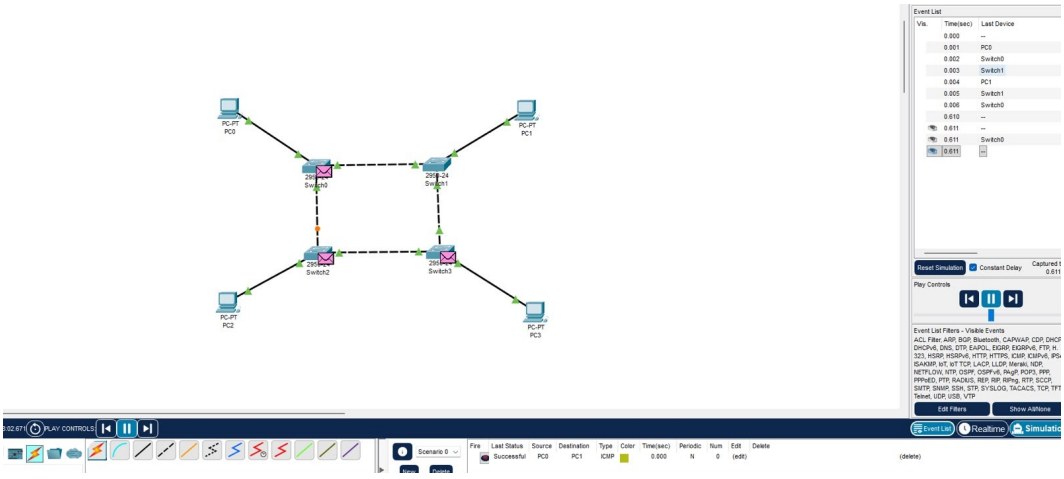


Fig. 5: Ring topology in Cisco Packet Tracer

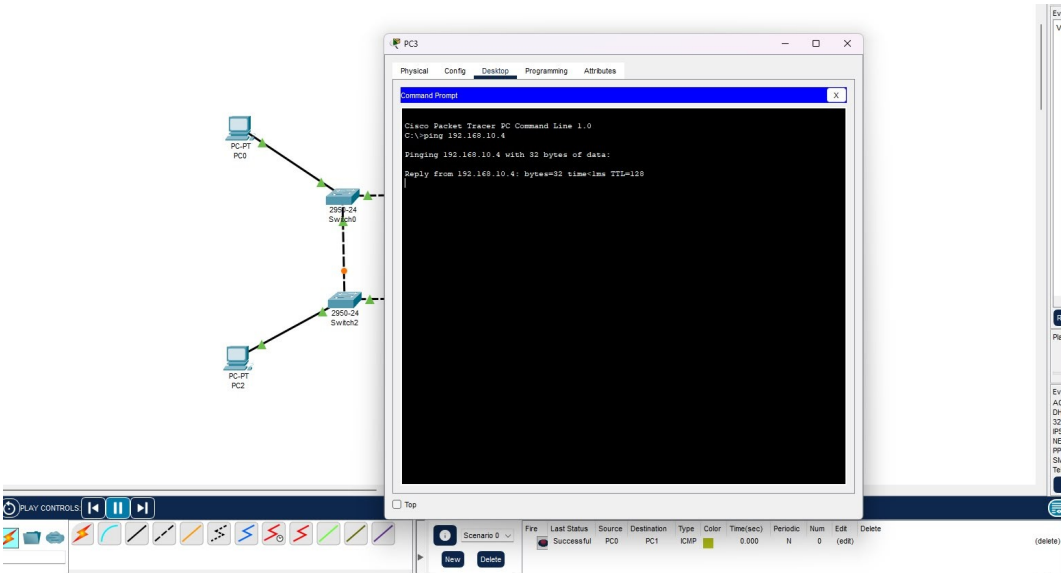


Fig. 6: Ping test between the PCs in the ring topology

5.4 Mesh Topology

In the mesh topology every computer was connected to every other computer with crossover cables, so each pair of devices has its own direct link. After assigning the IP addresses, the ping command was used to test all the paths.

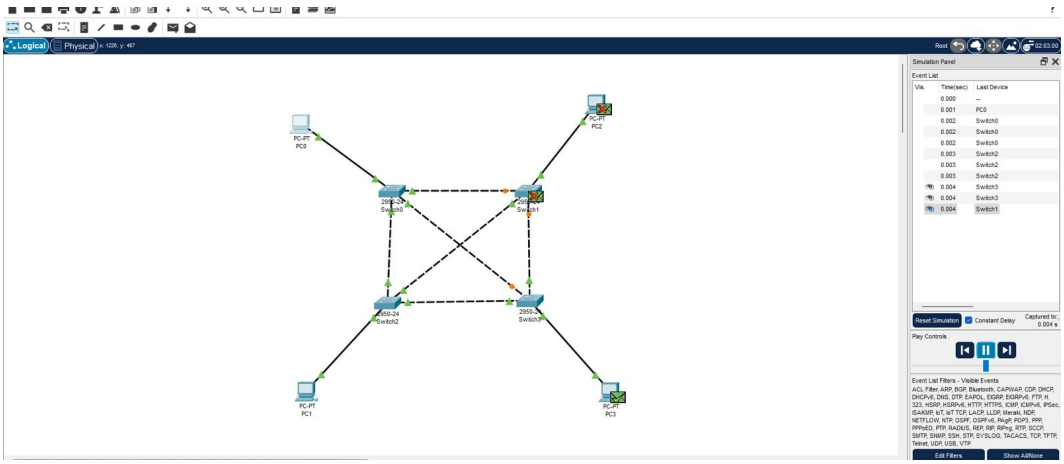


Fig. 7: Mesh topology in Cisco Packet Tracer

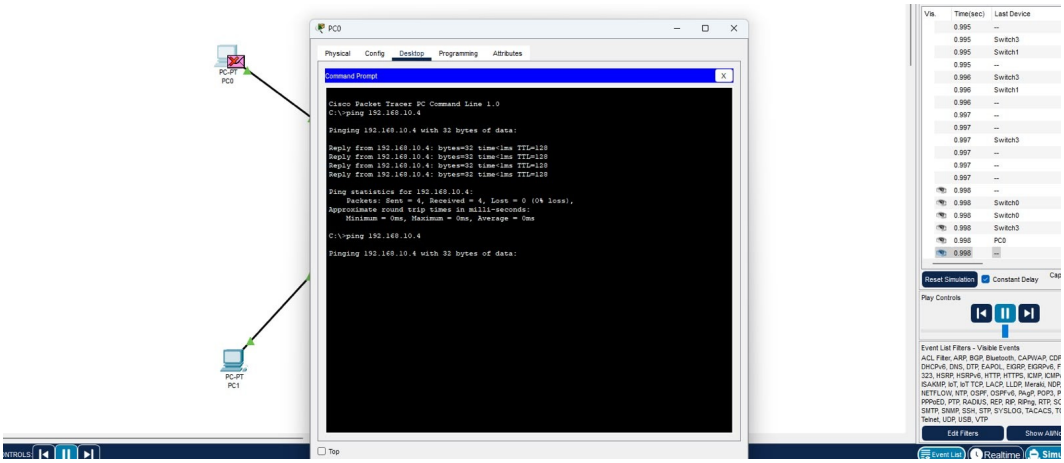


Fig. 8: Ping test between the PCs in the mesh topology

6. Result

The bus, star, ring and mesh topologies were successfully implemented in Cisco Packet Tracer with three computers each. After IP addresses were assigned, the ping command showed that the computers could communicate with each other in every topology.

7. Conclusion

This lab helped me understand how the different network topologies are arranged and how they are built in Packet Tracer. The bus topology is simple and cheap, but the single backbone is a weak point. The star topology is easy to manage because every device connects to a central switch. The ring topology connects devices in a loop, and the mesh topology provides direct links between all the devices, which makes it the most reliable but also the most expensive to cable. Testing with the ping command confirmed that each network was configured correctly.